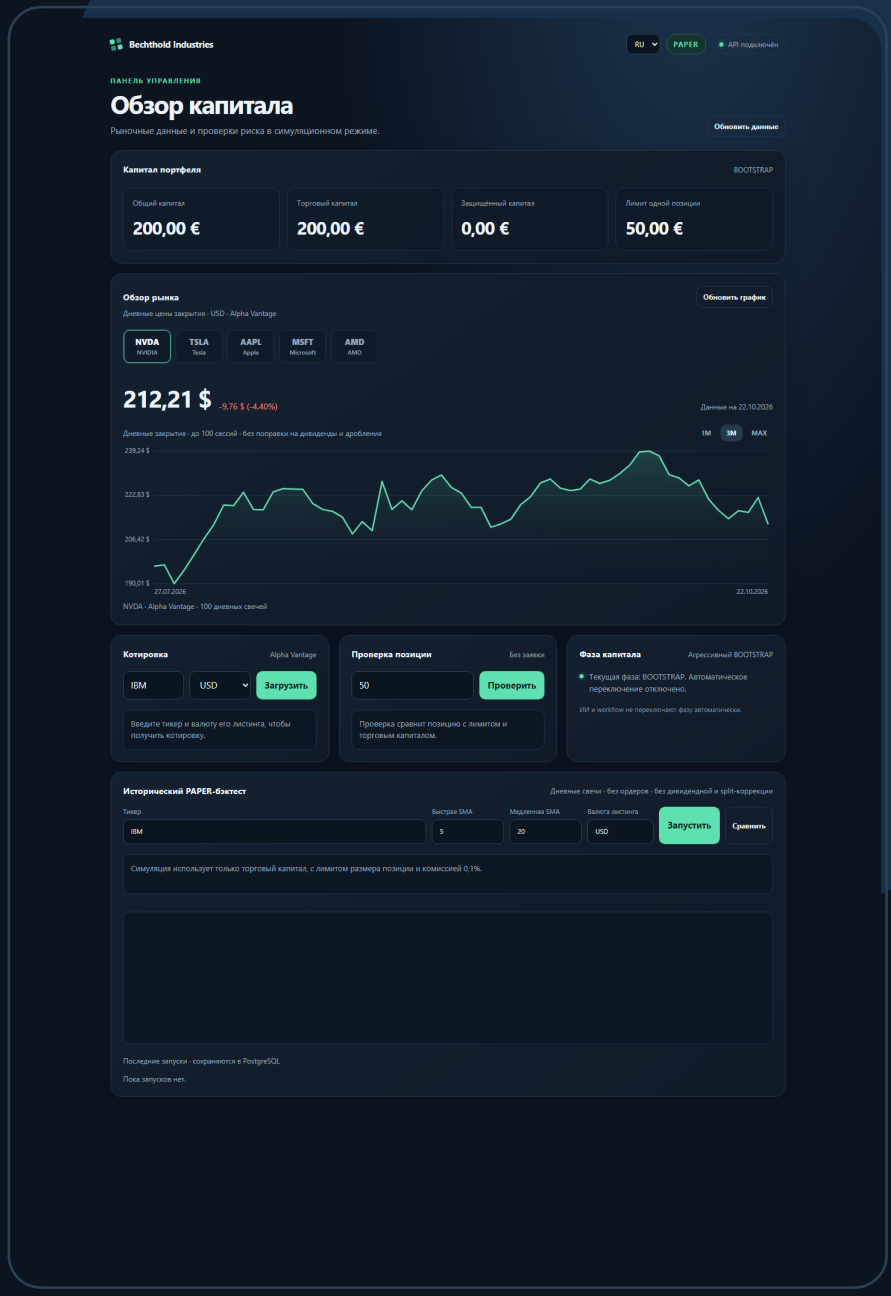
PRODUCT CASE STUDY / 2026

Market research, built around capital protection.

A local-first market dashboard for desktop and iPhone. It combines a curated stock watchlist, historical charts, risk checks and repeatable PAPER strategy analysis.

PYTHON / FASTAPI / POSTGRESQL / DOCKER / N8N



PROTECTED BY DESIGN

Capital buckets stay separate; changing phase requires a manual confirmation.

PAPER ONLY

No broker connection or real-money order execution path.

5 EQUITIES

NVDA, TSLA, AAPL, MSFT and AMD, with selectable daily charts.

3 LANGUAGES

Russian, German and English interface options.

PRODUCT DECISIONS

Useful analysis with hard boundaries

The product keeps the simulation understandable and makes protected money unavailable to strategy sizing.

Market watch

Select one of five widely followed US tickers. The chart shows up to 100 daily closes from Alpha Vantage, with 1M, 3M and MAX ranges.

Risk check

Every candidate is compared with the position cap and available trading capital. A failed check cannot flow into an approved result.

Strategy analysis

An SMA crossover backtest uses the next daily open for execution, applies estimated fees, and reports a separate later holdout period.

Installable web app

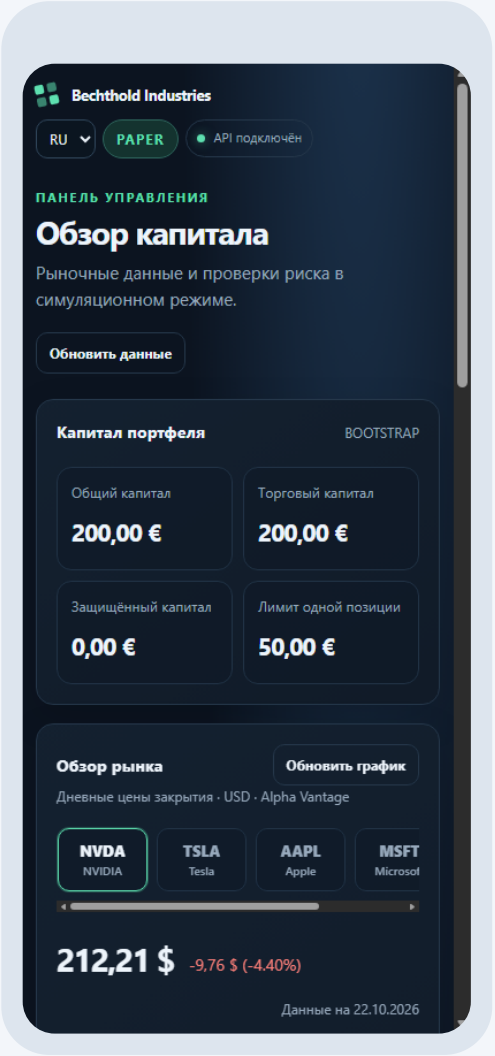
Responsive dashboard for desktop and iPhone home screens over the same private Wi-Fi. No public hosting is required.

CAPITAL MODEL

Start: 200 EUR trading capital

At 350 EUR equity: reserve 200 EUR

The BOOTSTRAP -> PROFIT_ONLY transition waits for a token-protected manual confirmation.



390 x 844 CSS px - responsive mobile preview

ENGINEERING AND VERIFICATION

A small system with explicit responsibilities

The API owns capital rules and analytics; PostgreSQL persists state; n8n coordinates only a PAPER workflow.



n8n orchestration: Strategy Engine -> risk gate -> Position Sizing -> explicit PAPER result

43 API tests

43 PASS

Capital protection, manual phase confirmation, market data validation, backtest timing and cost assumptions.

7 workflow tests

7 PASS

Inactive workflow, safe HOLD defaults, risk rejection behavior, and restricted result fields.

Continuous integration

CI GREEN

Latest GitHub Actions run passed both suites. Docker Compose configuration validates; API readiness and n8n health endpoints respond.

KNOWN LIMITS

Daily source data can be delayed, rate-limited and unadjusted for dividends or splits. Backtests omit slippage, taxes and other market effects. The app is for one trusted local user and is not hardened for public or multi-user hosting.

Source repository: github.com/Mafrik/bechthold-industries (private). Local install and screenshots are included in the repository.